BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Electric and Gas Service.

Case 25-E-XXXX, 25-G-XXXX

January 2025

Prepared Testimony of:

Cyber and Physical

Security Panel

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CONSOLIDATED EDISON COMPANY OF NEW YORK, INC.
DIRECT TESTIMONY OF CYBER AND PHYSICAL SECURITY PANEL

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CONSOLIDATED EDISON COMPANY OF NEW YORK, INC.
DIRECT TESTIMONY OF CYBER AND PHYSICAL SECURITY PANEL

I. INTRODUCTION

A. Introduction and Qualifications of Panel Members

Q. Would the members of the Cyber and Physical Security Panel (Security Panel or Panel) please state your names and business addresses?

A. Our names are Nicholas Boshears, Angel Cardoza, and Kurt John. Our business address is 4 Irving Place, New York, New York 10003.

Q. By whom are the Panel members employed?

A. We are all employed by Consolidated Edison Company of New York, Inc. ("Con Edison" or the "Company").

Q. Please explain your current role and responsibilities as well as your work history.

A. **(Boshears)** I am the Director of Corporate Security. In this role, I am responsible for managing all security-related investigations, tactical response situations, physical security assessments and mitigations, security-related capital projects, the Insider Threat program, and the Security Operations Centers for Con Edison and O&R. Prior to joining Con Edison in December 2023, I was employed with the Federal Bureau of Investigation (FBI) for 24 years, holding positions of increasing responsibility

focusing on domestic and international criminal, counterintelligence, and counterterrorism investigations. In my last assignment with the FBI, I served as Special Agent-in-Charge, New York Field Office, Operational Support Division, which included responsibility for the FBI's personnel security, crisis management, crisis intervention and response teams, information technology, internal affairs investigations, insider threat, and physical security programs in the New York City metropolitan area.

(Cardoza) I started my career in the Company in 1998. I am currently the Vice President of Substation Operations, but as of February 1, 2025, I will be the Vice President of System and Transmission Operations (STO). In Substations, I am responsible for the construction of new substations and for the operations and maintenance of 103 substations, which includes compliance with applicable standards. In STO, I will be responsible for several departments, including Transmission Planning, which plans the overall development of the electric bulk power transmission system, coordinating with the New York Independent System Operator (NYISO) and neighboring utilities; System Operations, which ensures the safe operation of our electric and steam

systems; Transmission Operations, which constructs, inspects, maintains, and repairs the underground and overhead transmission system; and Transmission Project Development, which coordinates the project portfolio of transmission investment.

My previous general manager positions at Con Edison include Protective Systems Testing in Substation Operations, and Gas Engineering. I was also a general manager in Electric Operations, the Control Center, and Substation Operations at Orange and Rockland Utilities, Inc. (O&R).

(John) I am a Senior Cybersecurity Technology Executive with extensive Cybersecurity experience in creating and scaling agile, resilient, and secure technology ecosystems. I joined Con Edison in October 2024 as Vice President and Chief Information Security Officer. I am responsible for governance, strategy, and operations of Cybersecurity across Con Edison. This includes the security of all corporate data, customer information, as well as our business operations (including gas, electric, and steam). Prior to Con Edison, I served as Chief Information Officer (CIO) and Chief Security Officer (CSO) at Expedia Group, where I oversaw enterprise IT, reliability and platform

engineering, physical security, cybersecurity, and privacy. Prior to Expedia, I was Chief Cybersecurity Officer for Siemens USA, where I managed product and information security for Siemens' $25 billion in United States business. Leading up to my role at Siemens, I spent 10 years performing technology and Cybersecurity consulting and advisory services across various industries.

Q. Please state your educational background.

A. **(Boshears)** In 1990, I graduated from Oregon State University with a Bachelor of Science degree. Additionally, in 2012, I graduated from the Marine Corps University with a Master of Science degree and am a Certified Fire and Explosion Investigator. I also completed the Carnegie Mellon University Insider Threat Program Manager Certificate course.

(Cardoza) In 1998, I graduated from the City College of New York with a Bachelor of Engineering degree in Electrical Engineering. In 2009, I graduated from Mercy College with a Master of Science degree in Organizational Leadership. In 2017, I graduated from Fordham University with a Master of Business Administration degree.

(John) In 2008, I graduated from University of the West Indies with a Bachelor of Science degree in Computer Science. Additionally, I am a Certified Information Security Professional.

B. Purpose of Filing

Q. Before summarizing the purpose of this testimony, please explain the Company's thoughts on security.

A. Maintaining both the cyber and physical security of our assets is paramount to the Company. We must be able to build and operate our systems safely and securely. On the cyber side, cybersecurity incident risk has increased dramatically in recent years, as multiple organizations have experienced impacts to their operations and lost confidential customer information. And attackers have become more sophisticated. In particular, the utility industry has seen a 70% increase in attacks from 2023 to 2024. (Cyberattacks on US utilities surged 70% this year, says Check Point, September 11, 2024, https://www.reuters.com/technology/cybersecurity/cyberattacks-us-utilities-surged-70-this-year-says-check-point-2024-09-11/). Consequently, we must continue to develop our capabilities and implement technology.

Q. Please explain the need for cybersecurity investments.

A. Cybersecurity investments must address current risk exposure. In addition to known everyday threats, emerging trends, regulations, and incident reports from peers, the Company also monitors emerging technologies that will drive cybersecurity investment, including:

- Application of Artificial Intelligence (AI) and Machine Learning (ML) for defensive and offensive cyber protection; and
- The continued implementation of the industry standard zero-trust security model(s).

Each of these indicators has the potential to alter the tools, resourcing, and architecture of cybersecurity. AI/ML capabilities are essential due to the growing complexity, sophistication, and volume of cyber threats. Traditional cybersecurity tools often struggle to keep pace with the dynamic nature of modern attacks, and AI/ML can enhance the detection, prevention, and response capabilities of these tools.

Unlike traditional security models that rely on perimeter defenses, zero trust assumes no entity can be trusted by default and that threats can come from both outside and

inside the network, and it treats every user, device, and network component as potentially untrusted until verified. Hardware and software technologies continue to advance and require additional functionality to mitigate these evolving cybersecurity risks.

Q. Please continue with the physical security side.

A. Physical security risks have also increased dramatically. Utilities throughout the Country continue to see attacks on physical assets, theft of property, and harm to personnel. Terrorism, whether domestic or foreign, remains a threat and the Department of Homeland Security's (DHS) Homeland Threat Assessment for 2025 indicates that the terrorism threat environment will remain "high."

In New York City, we recently experienced a CEO of a large corporation murdered in Manhattan. Workplace violence remains a concern, particularly with the threats and assaults committed by the public against our employees. The increasing level of displaced and unhoused individuals in New York City also causes those individuals to attempt to access facilities where they can stay warm (or cool) and sleep comfortably.

Despite local and federal regulations, drone activity near

our critical infrastructure remains increasingly high; however, no specific authority has been granted to mitigate this threat.

Throughout 2024, there was an increase in propaganda and rhetoric from ISIS and Al-Qaeda calling for the targeting of the U.S. and in particular, critical infrastructure.

Last, the threat posed by Nation State Actors and other foreign intelligence entities remains at an extremely high level due to the Company delivering energy to the financial markets, United Nations, other diplomatic facilities, and military installations. The National Counterintelligence Security Center stated this threat is at an unprecedented level in breadth, volume, and impact.

Q. Please summarize and briefly explain the purpose of the Panel's testimony.

A. We explain the Company's security philosophy and present the Company's proposed cyber and physical security projects and programs and their respective funding requirements. Specifically, this testimony covers the capital and O&M funding requirements for the Company's security programs and initiatives.

We also explain the potential for an update as we expect regulations may be issued for which we are not requesting funding since the actual requirements are currently unknown.

Q. Will these projects also affect O&R?

A. Many projects and programs discussed in our testimony are common to the Company's electric, gas, and steam businesses, and, in most cases, to O&R. The Company's Accounting Panel explains how costs are allocated across Con Edison' businesses and, where applicable, to O&R.

Q. What organizations have projects for this Security Panel?

A. Enterprise Cybersecurity, Corporate Security, STO (specifically, for the Energy Control Centers), and Substation Operations have included cyber and/or physical security projects/programs in this testimony.

Q. Please discuss the coordination among these organizations.

A. We coordinate cyber and physical security projects and programs internally and on an enterprise-wide basis.

Q. What period does your testimony cover?

A. Our testimony covers the 12-month period ending December 31, 2026 (Rate Year, Rate Year 1, or RY1), the 12-month period ending December 31, 2027 (Rate Year 2 or RY2), and

the 12-month period ending December 31, 2028 (Rate Year 3 or RY3). The Company has proposed a rate plan for RY1 only but, as the Accounting Panel explains, is interested in pursuing a multi-year rate plan in settlement discussions with Staff and interested parties. We have provided RY2 and RY3 data for the purpose of facilitating any such discussions.

Q. How have you arranged this testimony?

A. We explain the Company's Security projects and programs, which we divide into two sections:

- Cyber Security Projects and Programs; and
- Physical Security Projects and Programs.

Throughout the project and program discussions, the Panel explains any additional full time equivalent (FTE) positions needed to implement the projects and programs described in our testimony and execute on the Company's goals.

Q. Will you be introducing any exhibits as part of your testimony?

A. Yes, we will be introducing the following exhibits:

- Exhibit ___ (CPSP-1) titled "Cyber and Physical Security Capital Project/Program Summary";

- Exhibit ___ (CPSP-2) titled "Cyber and Physical Security O&M Program Change Summary";
- Exhibit ___ (CPSP-3) titled "Cyber and Physical Security FTE Summary"; and
- Exhibit ___ (CPSP-4) titled "Cyber and Physical Security White papers."

Q. Please explain what is reflected in Exhibit ___ (CPSP-1).

A. Exhibit ___ (CPSP-1) summarizes the Company's projected capital expenditures for Cyber and Physical Security projects and programs in RY1, RY2, and RY3. These capital expenditures are organized into the main areas shown on the exhibit and noted above.

Q. Please explain what is reflected in Exhibit ___ (CPSP-2).

A. Exhibit ___ (CPSP-2) summarizes the Company's projected incremental O&M expenses for Cyber and Physical Security projects and programs in RY1, RY2, and RY3, compared to the 12-month period ending on September 30, 2024 (Historic Year).

Q. Please explain what is reflected in Exhibit ___ (CPSP-3).

A. Exhibit ___ (CPSP-3) summarizes the FTE positions needed by the Company to implement the Cyber and Physical Security projects and programs presented in this testimony and the other accompanying exhibits.

Q. Please explain what is reflected in Exhibit ___ (CPSP-4).

A. Exhibit ___ (CPSP -4) presents detailed capital and/or O&M white papers associated with the Cyber and Physical Security projects and programs discussed in this testimony.

Q. Were these exhibits prepared under your direction and supervision?

A. Yes.

MARK FOR IDENTIFICATION AS EXHIBITS __ (CPSP-1), ___ (CPSP-2), ___ (CPSP-3) and __ (CPSP-4)

D. Background

Q. What is security and why is it important?

A. Cybersecurity is the governance, strategy, and operations associated with keeping our networks, data, and operations safe from cyberattacks and disruption. Physical security focuses on ensuring that our employees, contractors, visitors to company facilities, and assets remain protected and secure from physical attacks, whether emanating from an external threat or from an insider.

Cyber and physical security are critical for protecting the Company's personnel, facilities, operations, and data. These are core to maintaining the Company's viability and continued operations. It is also imperative for protecting the privacy of our employee and customer data, as well as

our Critical Energy/Electric Infrastructure Information (CEII). As a critical infrastructure company, Con Edison faces physical and cyber threats from both domestic and foreign adversaries. Continuing investment in our security programs is critical for keeping up with the increasing risks and threats.

Q. Please further explain what is meant by cybersecurity.

A. Cybersecurity is the protection of the Company's computing system, whether it be Informational Technology (IT) or Operational Technology (OT) or any other computing system. Cybersecurity risks and incidents continue to increase, as can be seen by multiple other organizations experiencing impacts to their operations and losing confidential customer information.

Q. What is meant by OT?

A. OT, or Operational Technology, refers to hardware and software that detects or causes changes through direct monitoring and control of physical devices, processes, and events in the enterprise.

Q. Does this testimony contain requests for both mandated regulatory projects as well as projects that will enhance our cyber or physical security posture?

A. Yes. This testimony addresses projects associated with IT, OT, North American Electric Reliability Corporation (NERC) Critical Infrastructure Protection (CIP), and the federal Transportation Security Administration (TSA) that are required for cyber and/or physical security purposes.

Q. Are there other cyber-related projects discussed elsewhere in this filing?

A. Projects involving updates or patches to maintain pace with manufacturer changes or new systems are covered in the IT Panel testimony.

Q. What is NERC?

A. NERC is a not-for-profit international regulatory authority whose mission is to ensure the effective and efficient reduction of risks to the reliability and security of the transmission system. NERC develops and enforces Reliability Standards; annually assesses seasonal and long-term reliability; monitors the bulk power system through system awareness; and educates, trains, and certifies industry personnel. NERC’s area of responsibility spans the continental United States, Canada, and the northern portion of Baja California, Mexico. NERC is the Electric Reliability Organization (ERO) for North America, subject

to oversight by the Federal Energy Regulatory Commission (FERC) and governmental authorities in Canada. NERC's jurisdiction includes users, owners, and operators of the bulk power system, which serves nearly 400 million people.

Q. What is North American Electric Reliability Corporation Critical Infrastructure Protection (NERC CIP)?

A. NERC CIP is a set of standards aimed at regulating, enforcing, monitoring, and managing the security of the Bulk Electric System (BES). These standards apply specifically to the cyber and physical aspects of protecting the BES. The CIP standards provide a cybersecurity as well as a physical security framework to identify and secure critical assets that can impact the efficient and reliable supply of electricity.

Q. What assets are covered by NERC CIP?

A. The BES, that is, in general, assets greater than or equal to 100 KV.

Q. What is a High Value Network (HVN) or Asset?

A. Con Edison has a segmented computing system, that is, higher value assets are not in the same area as lower value assets. A HVN is a network that contains critical technology needed to operate our electric, gas, and steam

infrastructure. It is critical to keep this infrastructure secure and reliable.

Q. Is a HVN the same as or different than a NERC CIP asset?

A. HVN includes NERC CIP assets as well as other critical distribution assets. Con Edison has critical assets used as part of service delivery that are not part of the BES but are considered HVN. For example, this would cover our distribution system.

Q. What is the TSA's role in regulating cyber and physical security?

A. After the 2021 Colonial Pipeline incident, the TSA issued security directives, covering cyber and physical requirements, to certain qualifying utilities with gas pipelines. The TSA subsequently issued several security directives and may issue permanent requirements in 2025.

Q. Please provide an example of TSA's requirements.

A. The TSA conducts a nationwide Corporate Security Review program for pipeline security. One requirement is the presence and review of the Company's physical security plan for natural gas pipelines as well as an incident response plan for cyber events. Following review of these plans, the TSA provides improvement recommendations.

Q. Please explain what is meant by physical security.

A. Physical Security includes essential equipment and practices designed to protect physical assets, facilities, and personnel from unauthorized access, theft, vandalism, and other physical threats. Components of physical security include access control, perimeter security, surveillance and monitoring, security personnel, physical barriers, security lighting, intrusion detection systems, and visitor management.

Q. Please explain the term Insider Threat.

A. Insider Threat describes the threat posed by employees, contractors, vendors, and visitors who may cause harm or damage through intentional or unintentional actions. The Company's Insider Threat Program works across all organizations in the enterprise, including IT, Cybersecurity, Human Resources, Supply Chain, and Finance, among others, to identify risks posed by insiders and mitigate potential physical and/or cyber threats as well as threats posed by external or foreign intelligence entities.

Q. What kind of work does Corporate Security perform?

A. Corporate Security is responsible for overall physical security at Company locations. This includes providing

access to facilities, performing vulnerability assessments at critical facilities, managing two 24x7 Security Operations Centers, reviewing footage from thousands of cameras, investigating incidents involving potential criminal conduct by employees or the public targeting Company employees and assets, communicating with federal, state and local partners, participating in interagency and industry exercises and drills, delivering travel security briefings, and coordinating counterintelligence defensive briefings to address insider threats.

Q. What are the Company's objectives and general considerations when evaluating and selecting Security investments?

A. The Company considers several factors when evaluating and selecting Security investments.

For any enterprise risk, including cyber and physical security, we review and align risks with our business strategy and vision planning process, and implement solutions, policies, and procedures to provide robust protection and uninterrupted operations. We also engage in interactions with advisors, vendors, and Company stakeholders to select the most optimal solutions for our

needs. This collaborative approach allows us to maintain the goal of staying ahead in the ever-evolving business and technology landscape and deliver the best outcomes for Con Edison.

II. PROJECT AND PROGRAM DISCUSSION

A. Summary Information

Q. How much capital investment is the Company proposing for Cyber and Physical Security projects and programs?

A. We are proposing capital investments of $133.1 million in RY1, $108.9 million in RY2, and $260.0 million in RY3.

Q. How much additional O&M is the Company proposing for Cyber and Physical Security projects and programs?

A. For O&M, we are proposing to spend $67.4 million in RY1, $79.1 million in RY2, and $85.5 million in RY3. Of these amounts, O&M program changes account for a $22.4 million increase in RY1, with increases of $11.8 million in RY2 and $6.4 million in RY3.

Q. How many incremental Full time Equivalents (FTEs) are being requested by the Panel?

A. The Panel is requesting a total of 104 incremental FTEs, consisting of 78 FTEs to support cyber security projects

and programs, and 26 FTEs to support physical security projects and programs.

B. Cyber Security Projects and Programs

Background

Q. Does the Company have a cybersecurity program?

A. Yes. The Company's strategy employs the defense-in-depth concept. As new risks are identified, and the capabilities of adversaries increase, the Company reassesses current security controls, implements new processes, and invests in new technologies to maintain its security posture and try to stay ahead of malicious actors. Risks include operating failures of control systems, damage to transmission and distribution (T&D) assets, the loss of sensitive data, and employee and public safety due to a cyberattack or unauthorized access to systems or data.

Q. Does the Company work with others regarding cybersecurity?

A. The Company engages in collaborative defense efforts with industry, federal, state, and local partners. We also work on and participate in local, regional, and national level cybersecurity drills. We benchmark with and share best practices with peers within and outside the industry. Our cybersecurity protection program relies on a Collective

Defense and Collective Response framework.

Q. How has the Company been addressing the cybersecurity challenge?

A. The Company continues to address cybersecurity in alignment with the National Institute of Standards and Technology (NIST) Cybersecurity Framework: (1) Identify; (2) Protect; (3) Detect; (4) Respond; and (5) Recover.

Q. What is meant by identify and protect?

A. Identification and protection aim to avoid attacks on our system and are achieved through risk management processes, architecture and security reviews, and implementation of multiple technologies at multiple security zones to prevent unauthorized access to our systems and data, and to prevent disruptions to our operations. In addition, the Company has segmented its network into multiple areas to minimize the likelihood of an attacker accessing all parts of our network, i.e., reducing the impact of a cyberattack.

Q. Turning to the third step, detection, what does the Company do?

A. The Company operates a 24x7 Cybersecurity Operations Center (CSOC), which monitors our computing network to detect threats, anomalies, and vulnerabilities. We also partner

with external entities that provide the Company with intelligence to help mitigate potential threats.

Q. Please explain the fourth and fifth cybersecurity areas: Respond and Recover.

A. There are multiple approaches to addressing this area. To respond and recover, we have plans and procedures to respond to cyberattacks and data breaches, which the Company drills regularly. We also have contracts with cyber forensic experts, who provide forensic analysis, as well as an in-house cyber forensic team. In the event of an incident, we will acquire the necessary resources to both respond and recover. We also have disaster recovery and business continuity plans.

Q. Is there more work to do in the cybersecurity area?

A. Yes. Given the significant rise in the sophistication, volume, and impact of cybersecurity threats, we must continue to grow and develop our capabilities, implement technology, and enhance processes to further protect our systems and data, and to improve detection, resiliency, and recoverability.

Q. How are you addressing the continued work?

A. To stay ahead of the current and evolving threats, we must have technology in place to prevent and detect threats and upgrade these technologies as the risks evolve. Staying ahead of the threats means continuing and enhancing the work discussed above. The Company will also continue to work with others, partnering with law enforcement across federal, state, and local agencies, and benchmarking best practices with our industry peers.

Potential for Updates

Q. Are there other potential regulatory and legislative requirements regarding cybersecurity that may be needed in the next three years?

A. Yes. At the federal level, there are several pending rulemakings. For example, the DHS is developing regulations to implement the 2021 Cyber Incident Reporting for Critical Infrastructure Act (CIRCIA), which requires reporting for incidents within 72 hours; DHS though TSA has a pending rulemaking regarding gas pipelines cyber and physical security issues; and DHS through the Coast Guard also has a pending rulemaking. The Department of Treasury also sought comments as to whether the federal government should be the cyber insurance backstop.

FERC issued an order creating CIP 15 in 2024, requiring utilities to address internal network security monitoring for high and certain medium facilities. At the same time, FERC stated that the CIP requirement did not go far enough and NERC is developing a rule for endpoint monitoring. Recently, on January 17, 2025, the National Association of Regulatory Commissioners (NARUC) issued Cyber Distribution Baselines for utilities with the intent that States would implement these baselines.

Q. What about actions at the state level?

A. The New York State Public Service Commission (PSC or Commission) recently introduced a proceeding intended to review both IT and OT protections and perhaps mandate requirements.

Q. Please continue.

A. It is also important to note that at the end of the prior Trump administration, there were two executive Orders relating to purchasing equipment from certain nations, which may have required utilities to "rip and replace" existing equipment. It is possible that similar orders may be issued in the new administration.

Q. Does this filing address any of these costs?

A. No. Given that these requirements are new and emerging, the Company does not know the costs yet.

Q. Will the Company provide additional information regarding any cyber related regulations or laws?

A. The Company will update any information on these items as they become available, which may come after update. We will address any available information at update.

Projects and Programs

Q. On an overall basis, what is the Company's forecasted capital request for the cybersecurity projects and programs.

A. To continue enhancing the Company's security posture, the Company will invest in multiple cybersecurity efforts to secure both the IT and OT (i.e., segments of our network that support operational equipment, such as substations and electrical switches). The total RY1 capital request is $47.6 million; RY2 is $47.8 million; and RY3 is $47.9 million, for a total of $143.2 million.

Q. Please describe the forecasted O&M request for each rate year under the cybersecurity projects and programs.

A. The Company requests incremental O&M funding of $17.7 million in RY1, $9.7 million in RY2, and $3.6 million in RY3, totaling $31 million.

Q. Please describe total FTE request for the cybersecurity projects and programs.

A. There are 78 FTEs for the cybersecurity projects.

Q. Please breakdown the cybersecurity projects the Company plans to undertake.

A. The Company proposes investments in the following areas:

- Cybersecurity, requesting capital funding of $17.1 million in RY1, $17.2 million in RY2, $17.8 million in RY3, and incremental O&M funding of $14.4 million in RY1, $7.1 million in RY2, and $3.0 million in RY3, which includes 46 FTEs.

- Cybersecurity Infrastructure (Cyber for High Value Networks (HVN)), requesting capital funding of $5.8 million in RY1, $6.9 million in RY2, $7.0 million in RY3 and incremental O&M funding of $2.4 million in RY1, $2.5 million in RY2, and $0.6 million in RY3, which includes 20 FTEs.

- System and Transmission Operations (S&TO) Cyber Security and NERC Compliance, requesting capital funding of $7.2 million in RY1, $6.2 million in RY2, $5.7 million in RY3, and incremental O&M funding of $0.9 million in RY1, and $0.1 million in RY2, which includes 10 FTEs.

- Substation Critical Infrastructure Protection (CIP) NERC Cyber Security compliance, requesting capital funding of $15 million in each of RY1-RY3.

- Transmission Network Visualization System (TNVS), requesting capital funding of $2.5 million in each of RY1-RY3, which includes 2 FTEs.

Q. Do the first two projects – Cybersecurity and Cybersecurity for HVNs go together?

A. Yes. These programs are managed by the Enterprise Cybersecurity team and are intended to protect our HVN environment.

Q. Please explain the request for the Cybersecurity program.

A. There are projects needed at the corporate level to protect computing systems and networks. These programs use a risk-based approach and aim to reduce the risk in areas which continue to see an increase in both scale and velocity of attacks. Some key aspects of the program include:

- Cloud security enhancements to combine various security capabilities, such as network protection, zero trust and others into a single cloud-delivered platform, which is important as we continue to shift more programs and data into the Cloud;
- An advanced defense initiative to discover and visualize potential attack paths, helping us to better protect our

information assets and respond more quickly in the event of a breach; and

- Continue to identify and isolate the applications requiring the highest levels of security and protection so they are supported by only trusted individuals.

Q. Is this the complete list of work to be done under these two projects?

A. No. Other areas of focus include further securing applications used to operate our business; managing vulnerabilities; detecting unauthorized activity and incident response; protecting data, including personally identifiable information; securing our overall network environments; and refreshing software and hardware that are coming to their end-of-life.

Q. Please explain the Cyber for HVN program.

A. This program aims to address cybersecurity of our HVNs, ensuring that we keep the most sensitive aspects of our operations protected. Key projects include:

- Expanding Password Management programs, such as Privileged Access Management (PAM);
- Expanding application controls in the HVN to protect systems and data by allowing only approved, trusted

applications to run while blocking all other potentially harmful software;

- Enhancing HVN Endpoint Detection and Response (EDR) to improve our ability to protect end-user devices and increase our ability to detect and respond to malicious activity; and
- Enhancing asset and vulnerability management to aggregate asset information and associated vulnerability details to allow for better prioritization and faster response within our HVNs.

Q. Please explain the enhancements from these projects.

A. Overall, as described, these projects (Cybersecurity and Cyber for HVN) enable us to develop enhanced capabilities, technology, and processes to protect our systems and data, and improve detection, resiliency, and recoverability, and stay ahead of the current and evolving cybersecurity threats.

Q. Does the Company need to replace end-of-life hardware and software for these two programs?

A. Yes. We need to replace end-of-life hardware and software to benefit from the latest advancements in technology, maintain high levels of security, and ensure the

reliability and availability of our network, business applications, and infrastructure.

Q. Please explain the incremental O&M costs associated with these programs.

A. The O&M increase is primarily attributable to subscription and maintenance costs for these technologies. As noted by the IT Panel, the costs for maintaining purchased technology are growing. The Company's purchased technology has an annual subscription requirement and associated cost. Examples of increased subscription costs include cloud security enhancement subscriptions and costs for tools such as our Security Information and Event Management (SIEM) tool. In addition, we face costs associated with required contractor services for cybersecurity operations automation, technology activities support, and security testing and assessments.

Q. Please discuss the request for FTEs for both Cybersecurity programs.

A. To address evolving risks and continue improving and elevating the program, the Company has hired and continues to onboard talent to help address the increasing threat landscape. During the upcoming rate years, the Company

intends to increase Enterprise Cybersecurity headcount by 66 incremental FTEs, consisting of 46 FTEs to support our Cybersecurity program and 20 FTEs to support our Cybersecurity Infrastructure or Cyber for HVN program. Examples of some of the requested positions include:

- Improving the cybersecurity program by hiring five new FTEs at the Vice President and Director levels to provide enhanced senior level oversight and expertise in this challenging threat environment;
- Developing a dedicated HVN team to help improve our Cybersecurity posture, including Cyber operations, vulnerability management within the HVN, and improved identity and access management, comprising 20 FTEs;
- Enhancing the vulnerability management operations team to leverage best-in-class technologies and proactively identify vulnerabilities, assess the risks they present, and work with other stakeholders to mitigate these risks, comprising 5 FTEs; and
- Developing a dedicated application security and governance team to oversee effective security measures that help address rising security threats to software applications, comprising 2 FTEs.

The staffing increases proposed will improve our cyber security protections, increasing our maturity level.

Q. Will these enhancements guarantee that the Company will not face a cyberattack or incident?

A. Unfortunately, there are no guarantees regarding cyber-risks. The opponents we face are vast and, in many cases, are well funded, including some by nation states. Our goal is to minimize the likelihood of an attack as well as the potential impact in the event of such an attack. This work will enhance our posture and confirm that we are performing our due diligence and due care; however, our adversaries, like the Company, are always improving.

Q. Turning to the next project, what is the STO Cyber NERC Compliance Program?

A. This program improves the cybersecurity of the Energy Control Center and energy management systems to prevent intrusions, and to further comply with NERC standards such as CIP 3, 5, 10.

Q. What are the requirements associated with NERC CIP 3, 5, and 10?

A. CIP 3 deals with security management controls around protecting BES Cyber Systems. We address this with patch

management, networking port security, and firewall configuration management solutions to further increase our capabilities.

CIP 5 covers managing electronic access to BES Cyber Systems in support of an Electronic Security Perimeter. The proposed networking port security solution will help meet this CIP requirement.

CIP 10 covers configuration change management and vulnerability assessments. The patch management solution will help meet this standard.

Q. How is NERC CIP cyber addressed at the Control Center now?

A. Due to the separation requirement for NERC CIP systems, STO has three Cybersecurity teams: Threat Management, Network Security, and Vulnerability Management. The Control Center also has a dedicated Cyber Security Operations Center (SOC), which will continue to monitor network traffic and potential threats posed against the ECC 24/7/365.

Q. What will this program do?

A. This program expands the ECC's protections by implementing existing corporate technology, such as a network security platform, user endpoint behavior analytics, and load balancers, within the segregated ECC. Additionally, the

program will allow us to fully staff the ECC SOC on a 24/7/365 basis.

Q. What are the benefits of this program?

A. This program allows for more robust cybersecurity capabilities and protections.

Q. Are there any FTEs associated with this program?

A. There are ten FTEs associated with this program.

Q. What work will the FTEs do?

A. Seven FTEs will staff the dedicated Cybersecurity Operations Center focused on the security of the ECC network.

Two FTEs are for our Threat Management group and expanding existing technologies, such as our SIEM, Multi Factor Authentication, and Anti-Virus. They will also be working on our planned User Endpoint Behavior Analytics solution. Two other FTEs are for Network Security to help implement our network port security solution, and firewall configuration management tool. They will also assist in replacing our firewall hardware and will support other capital efforts within STO, such as firewall policies.

Q. Please explain the Critical Infrastructure (NERC) Security Upgrades program for Substations.

A. Because Verizon is changing its communication protocols, the Company needs to upgrade its Substation communication protocols from "serial" communications to routable communications. (Serial refers to "point-to-point" communication, whereas routable refers to "bi-directional" communication). This protocol change requires the Company to meet NERC compliance requirements for substations with External Routable Connectivity (ERC). These requirements include over 100 requirements across Medium and Low substations, such as placing firewalls at every location and having situational awareness using various security requirements.

Q. What will this project do?

A. This project will allow the Company to meet the NERC Compliance ERC requirements. This will include implementing, for example, new cyber security tools, servers, firewalls, password management, and role-based access.

Q. What is the final cybersecurity project you will discuss?

A. The Transmission Network Visualization System (TNVS).

Q. What is this system and what does it do?

A. TNVS is an application designed to provide situational awareness and provides backup capability for the Energy Management System (EMS). TNVS provides operators with tools to monitor and manage the transmission network even under the most adverse conditions.

Q. Please explain the project being requested.

A. We plan to add Data Diode servers to the TNVS applications, along with the established Disturbance Monitoring Equipment (DME) servers.

Q. Before we continue, please explain what a data diode is?

A. Data diodes are hardware-based devices designed to provide unidirectional data flow, meaning data can only travel in one direction. This technology prevents data from flowing back into the source network.

Q. Please continue.

A. The project objectives are to:

- Establish connectivity from Data Diode servers to the TNVS substation configuration.
- Implement a process to send information files from Data Diode servers to the TNVS server.
- Develop software and data structures on the TNVS server to receive and process information from Data Diodes.

Q. What are the benefits of this project?

A. Increased safety, reliability, and customer satisfaction: The TNVS project aims to enhance situational awareness and reduce the likelihood of human error, enhancing the cybersecurity and physical security measures of the TNVS application and network.

Q. Are any FTEs needed?

A. Two FTEs are being requested for the TNVS project.

Q. Please discuss the work the FTEs will do?

A. The FTEs will be responsible for overseeing the installation and integration of necessary equipment. This includes firewalls, modems, media converters, and other resources across all 40 transmission and 63 sub-transmission substations.

C. Physical Security Projects and Programs

Background

Q. What are Physical Security projects and programs?

A. Physical Security projects and programs focus on providing barriers, controls, and systems to protect and defend against physical attacks targeting the Company's facilities, infrastructure, employees, supply chain, and operations.

Q. How do physical Security projects and programs create value for customers?

A. Physical Security projects and programs protect the people and assets that deliver safe and reliable energy service to Con Edison's customers. Such projects and programs also help mitigate potential security risks and protect sensitive data including customer information. In these ways, Physical Security projects and programs create value for Con Edison's customers.

Q. How does Corporate Security manage and address Con Edison's physical security risks and needs?

A. Corporate Security implements projects and programs designed to address and mitigate Con Edison's evolving threat landscape including physical security risks, while meeting the needs of the Company's operating departments. Corporate Security uses a "defense-in-depth" strategy to deter, deny, detect, and delay potential threats. The first objective is to deter any type of attack. The second objective is to deny adversaries access to the targeted assets. Where the first two strategies fail, the third objective is to quickly detect the attack or situation, using surveillance, intrusion detection systems, human

observation, and/or technology that identifies inconsistencies, such as perimeter breaches. Once an intrusion or attack is attempted or in progress, the fourth objective is to delay the perpetrator using physical security and targeted hardening methods. Corporate Security personnel perform vulnerability assessments and leverage NERC-CIP, TSA, and industry best practices to address identified vulnerabilities, and respond to alarms and incidents.

Q. Is Corporate Security the only organization within Con Edison that addresses physical security risks and needs?

A. No. As with cyber security, STO, Substations, and Facilities play an important role in addressing physical security risks and needs, including through compliance with NERC CIP and PSC physical security recommendations. Corporate Security, STO, Substations, and Facilities work together to design and implement coordinated physical security improvements. The projects and programs discussed below reflect the Company's plan to continue synchronizing physical security priorities across the enterprise.

Funding Request

Q. What is the Company's total projected capital investment in Physical Security projects and programs for the three rate years?

A. We are proposing a total capital investment of $358.8 million for the three rate years. This amount includes capital investments of $85.5 million in RY1, $61.1 million in RY2, and $212.2 million in RY3.

Q. What is the Company's total projected funding need for O&M program changes related to Physical Security projects and programs for the three rate years?

A. We are proposing incremental O&M spending of $9.5 million across the three rate years. This amount includes incremental O&M spending of $4.6 million in RY1, $2.0 million in RY2 and $2.8 million in RY3.

Q. How many FTEs are associated with these requests?

A. The Company is requesting 26 FTEs to support its Physical Security projects and programs.

Projects and Programs

Q. What are some of the Physical Security projects and programs that the Company will undertake?

A. The Company plans to undertake the following Physical Security projects and programs during the rate years:

- **Protective Intelligence and Countermeasures**, with a total capital investment of $19.5 million, consisting of $5.5 million in RY1, $4.8 million in RY2, and $9.2 million in RY3, as well as incremental O&M spending of $4.0 million in RY1, $1.6 million in RY2, and $2.7 million in RY3.

- **Perimeter Enhancements**, with a total capital investment of $9.0 million, consisting of $3.0 million in each of RY1, RY2, and RY3.

- **Substation Physical Security Enhancements**, with a total capital investment of $30.0 million, consisting of $10.0 million in each of RY1, RY2, and RY3.

- **ECC Physical Security Enhancements**, with a total capital investment of $14.7 million, consisting of $5.8 million in RY1, $4.5 million in RY2, and $4.4 million in RY3, as well as incremental O&M spending of $1.4 million in RY1, $400,000 in RY2, and $600,000 in RY3.

- **ECC Relocation**, with a total capital investment of $265 million, consisting of $55 million in RY1, $35 million in RY2, and $180 million in RY3.

Q. Please describe the Protective Intelligence and Countermeasures program.

A. The Protective Intelligence and Countermeasures Program will align the Company's Physical Security program with the DHS Cyber Security and Infrastructure Security Agency's (CISA) guidance for assessing, deterring, detecting, and delaying threats.

Q. What do CISA's guidelines suggest?

A. The DHS/CISA Office of Bombing Prevention guidelines focus on performing Random Anti-Terrorism Measures, which provide

for methodical deployment of security resources to potential facility targets. Such measures involve varied levels of screening, surveillance, and detection capabilities above and beyond permanent security systems.

Q. Please provide examples of measures the Company will put in place to address the guidelines.

A. The Company intends to address many of the measures suggested by the DHS/CISA Office of Bombing Prevention. For example, we intend to implement Random Anti-Terrorism Measures by randomly deploying countermeasures at different Company locations.

As another example, the Company plans to improve its incident investigation methodology by implementing new case management software, which will provide for improved data integration and analysis, including threat detection information. This new software will also help enhance collaboration between Investigations, Insider Threat, Executive Protection, Physical Security, and Business Ethics and Compliance.

Finally, we plan to build out our Insider Threat program to strengthen the Company’s ability to identify associations and affiliations with adversaries targeting our critical

infrastructure. These improvements, which will help the Company meet the CISA guidelines and recommendations, include the use of software that will help us identify threats posed by Nation State Actors targeting our employees, data, and systems.

These items, and the others described in the program white paper, presented in Exhibit ___ (CPSP-4), will provide an enhanced physical security program during these times of heightened threats.

Q. Are you proposing to hire additional FTEs to support this program?

A. Yes. As part of the Protective Intelligence and Countermeasures program, the Company is proposing to hire 23 incremental FTEs by the end of RY1 to enhance its Threat Intelligence platform, Insider Threat program, Random Anti-Terrorism Measures, Investigations/Tactical Response Team Investigations, and Executive Protection program. These 23 incremental FTEs will be allocated as follows:

- 6 to support the Insider Threat program;
- 4 to support the Executive Protection program;
- 2 to support Random Anti-Terrorism Measures;
- 4 to support Facilities Security;

- 5 to support Investigations and Tactical Response; and
- 2 to support Supply Chain.

Q. Please explain why current staffing levels are insufficient.

A. Current staffing levels are inadequate due to contractor attrition and turnover in the Security Operations Center, work volume assigned to investigators, requirements for analytical and counterintelligence expertise focused on Insider Threat matters, and protective screening duties to comply with Con Edison policy and New York State laws.

Q. How did you arrive at these required staffing levels?

A. These staffing levels are based on turnover rate within the SOC as well as current and anticipated work requirements to support the Protective Intelligence and Countermeasures Program, including Insider Threat, Executive Protection, and Random Anti-Terrorism Measures, during the rate years.

Q. Does the Company's funding request include O&M program changes for the 23 incremental FTEs discussed above?

A. No. The Company's funding request includes O&M program changes for 14 of the 23 incremental FTEs discussed above. The details including funding requirements for the other nine FTEs were not developed in time to be included in this

filing. For this reason, the Company plans to present the additional FTE details and funding requirements in its update filing.

Q. Are you submitting any exhibits that detail the Protective Intelligence and Countermeasures program?

A. Yes. We have prepared a detailed white paper discussing this program. This white paper is titled "Protective Intelligence & Countermeasures Program" and is provided in Exhibit ___ (CPSP-4).

Q. Please describe the Perimeter Enhancement program.

A. The Perimeter Enhancement program focuses on strengthening layers of protection or "barriers" surrounding Company facilities to deter, detect, deny, delay, and defend against potential physical intrusions. Such "barriers" include physical fences and gates, and around-the-clock security guards.

Q. Please describe what the Company is proposing with its Perimeter Enhancement program.

A. The Company is proposing to replace older 2-inch x 2-inch mesh fencing with enhanced high-security hardened fencing.

Q. Why is this program needed?

A. Recent experiences at Con Edison and other utilities have demonstrated that the existing 2-inch x 2-inch mesh fencing is no longer an effective security barrier, although such fencing does generally delay intrusions. The enhanced hardened fencing provides a more effective deterrent and delays potential intrusions.

Q. Are you submitting any exhibits that detail the Perimeter Enhancement program?

A. Yes. We have prepared a detailed white paper discussing this program. This white paper is titled "Corporate Security Perimeter Enhancement" and is provided in Exhibit ___ (CPSP-4).

Q. Please describe the Substation Physical Security Enhancement Program.

A. The Substation Physical Security Enhancements program focuses on upgrading fencing and camera systems at several Con Edison substations, including, for example, the Jamaica, East 179th Street, Dunwoodie, and Sprain Brook substations.

Q. What are the benefits of this program?

A. The Substation Physical Security Enhancements program will install new anti-climb fencing and cameras equipped with AI

technology and intrusion detection capabilities. Both efforts will help protect our perimeter.

Q. Turning to the next program, please describe the Energy Control Center (ECC) Security Enhancements Program.

A. The ECC Security Enhancements Program provides support for the uninterrupted operation of the Company's bulk power control centers, consisting of the ECC and the Alternate Energy Control Center (AECC). For these facilities, uninterrupted operation is essential for maintaining the reliable and safe operation of the BES and Steam System. This involves compliance with regulatory requirements including NERC CIP standards and other requirements from the New York State Reliability Council (NYSRC) and the Northeast Power Coordinating Council, Inc. (NPCC).

Q. What will this program do?

A. This project will focus on:

- Enhancing emergency evacuation systems at control centers.
- Implementing detection systems for chemical and biological threats in HVAC systems.
- Continuing implementation of drone detection systems.
- Upgrading card access control systems and

physical access control systems.

- Replacing battery systems for ECC UPS units to ensure uninterrupted power supply.

Q. What are the benefits of this program?

A. This program will enhance security of the ECC so that we continue to keep pace with technology. These enhancements are needed at the existing ECC for regulatory compliance with NERC CIP.

Q. Is the Company requesting additional FTEs to support this program?

A. Yes. The Company is requesting two incremental FTEs to support this program.

Q. What will these incremental FTEs do?

A. These incremental FTEs will be dedicated to managing essential security functions, including card access systems, video analytics, security cameras, drone detection equipment, biometric screening systems, and other critical tasks.

These roles will address emergent security needs and bolster monitoring capabilities within the control centers, thereby maintaining high levels of security and operational reliability for the BES and Steam System.

Q. Before turning to the next project, ECC Relocation Project, please explain why are there two ECC projects?

A. One project focuses on current enhancements, while the other is a long-term project to relocate the ECC, which will take approximately a decade.

Q. Please describe the last project, the ECC Relocation Project.

A. Con Edison's ECC is in Manhattan, a densely populated area no longer suitable for a modern BES control center. The location is adjacent to several high-risk targets and above a substation.

To address these challenges, Con Edison plans to relocate the ECC to a new location, where the Company can incorporate industry best practices. These best practices will include, for example, having a 300-foot perimeter fence setback and a winding entrance road. This project is an investment designed to strengthen the Company's energy control infrastructure against potential non-climate hazards and vulnerabilities. This project will address and mitigate the physical security risks associated with the existing ECC.

Q. What are the benefits of this project?

A. The new ECC will provide for reliable and resilient operation and monitoring of the BES, particularly including areas with critical infrastructure such as the financial markets, airports, transportation hubs, and hospitals.

Q. What level of capital funding is needed to support this project's lifecycle?

A. This project has an estimated timeline of ten years, covering three rate cases. The total as for this project is estimated at $1.1 billion.

Q. What level of capital funding is needed to support this project during the three rate years?

A. We are proposing a total capital investment of $270 million during the three upcoming rate years. This amount includes proposed capital investments of $55 million in RY1, $35 million in RY2, and $180 million in RY3. These amounts are related to consultants, design work, and property purchase(s).

Q. Are there any other Physical Security projects or programs in which the Company plans to make capital investments during the upcoming rate years?

A. Yes. The Company also plans to make capital investments in the three additional Physical Security programs summarized

below. Additional information regarding these projects and programs can be found in the white papers provided in Exhibit ___ (CPSP-4).

- **Corporate Security Technology and Equipment Enhancement program** (also referred to as the Network Video Recorder and Digital Video Recorder Replacement program), which will enhance security technology and replace equipment that has reached its end-of-life. For this project, the Company is requesting capital funding of $1.71 million in RY1, and $1.5 million in each of RY2 and RY3, as well as incremental O&M funding of $860,000 in RY1, and $750,000 in each of RY2 and RY3. The Company is requesting 1 FTE to support this program. This FTE is needed to assist with the operational responsibilities for our Card Access system.

- **Company-Wide Camera Rollout program**, which will replace legacy cameras with new digital cameras at all Company locations. For this program, the Company is requesting capital funding of $1.688 million in RY1, and $1.5 million in each of RY2 and RY3.

- **Facilities Security Upgrade program**, which will involve physical security upgrades and enhancements to various buildings and yards managed by Facilities. For this project/program, the Company is requesting capital funding of $4 million in RY1, and $3 million in each of RY2 and RY3.

Q. Does this conclude the Panel's direct testimony?

A. Yes, it does.